



ENGINEERING SPECIFICATIONS

Document Name: Konnect Installation Manual		
Document No: ESP-19-20-008	Revision: 00	Date:29/01/2020
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QFlash User Guide

Rev. QFlash_User_Guide_V3.4

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Firmware Upgrade Procedures

The firmware can be upgraded through the following three steps by the QFlash tool.

Step 1: Set serial port and baud rate.

Step 2: Load firmware files.

Step 3: Upgrade the firmware.

The following describes the details of how to use the tool to upgrade firmware.

1.1. Configure Serial Port and Baud Rate

When the QFlash tool is opened, the main interface will show as below.

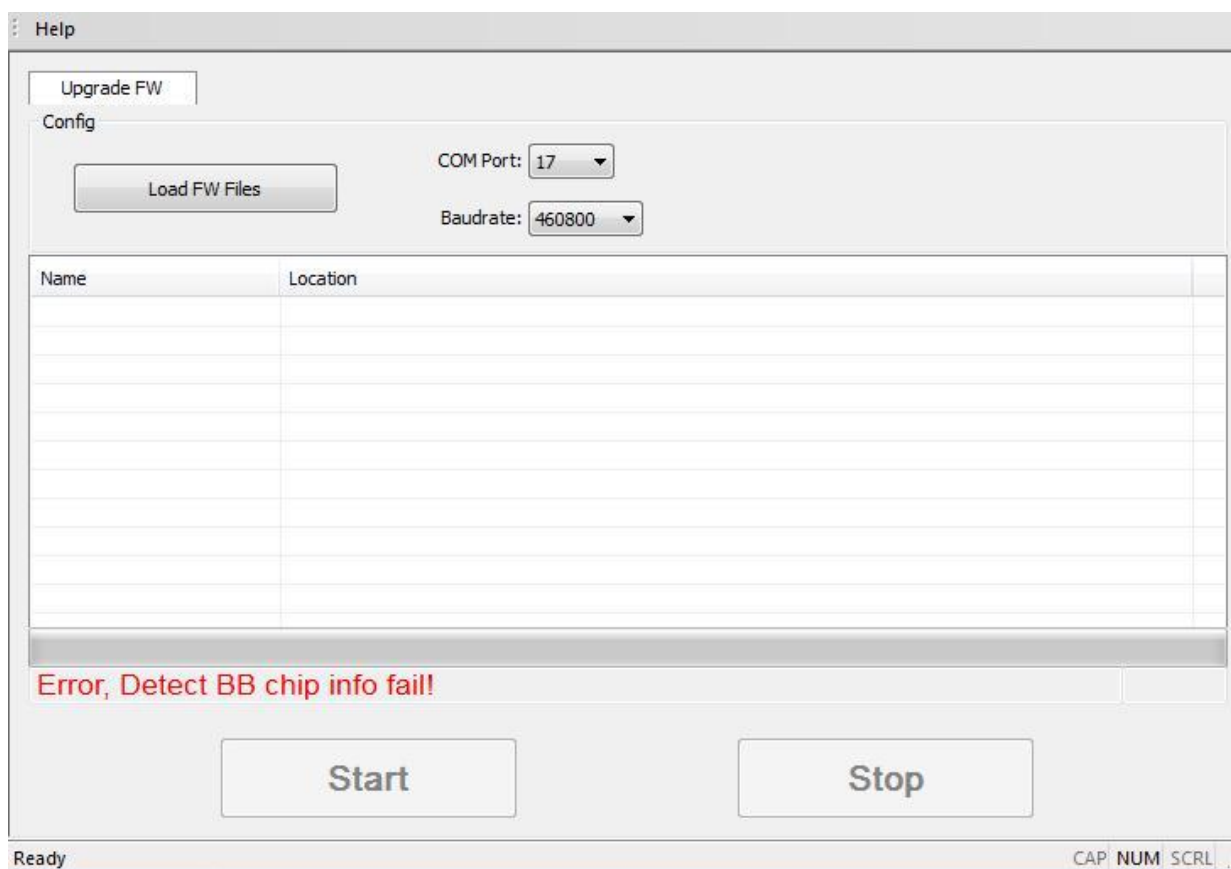


Figure 1: Main Interface

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1.1.1. Set Serial Port**1.1.1.1. COM Port Selection for M66 Modules**

Click “**COM Port**” dropdown list to select the COM port through which the firmware is upgraded. As shown in the following figure.

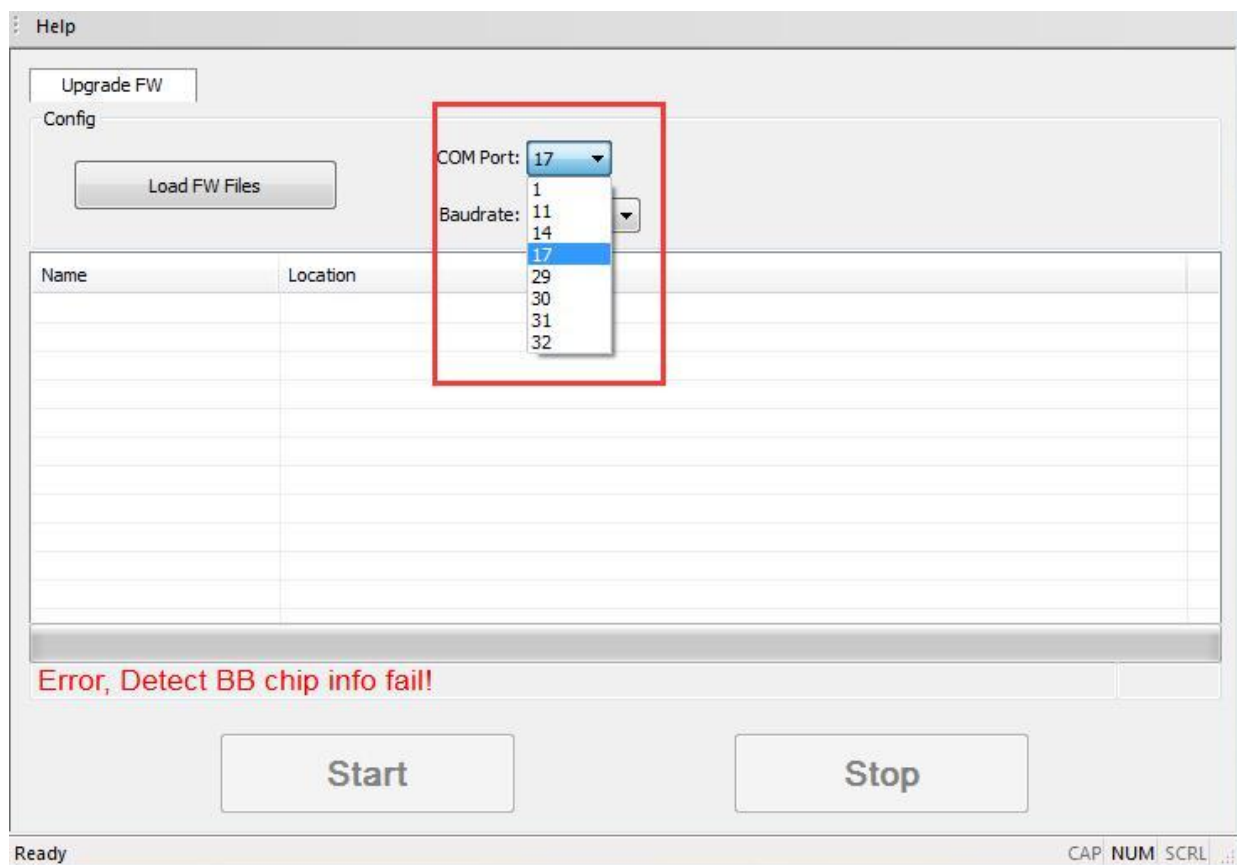


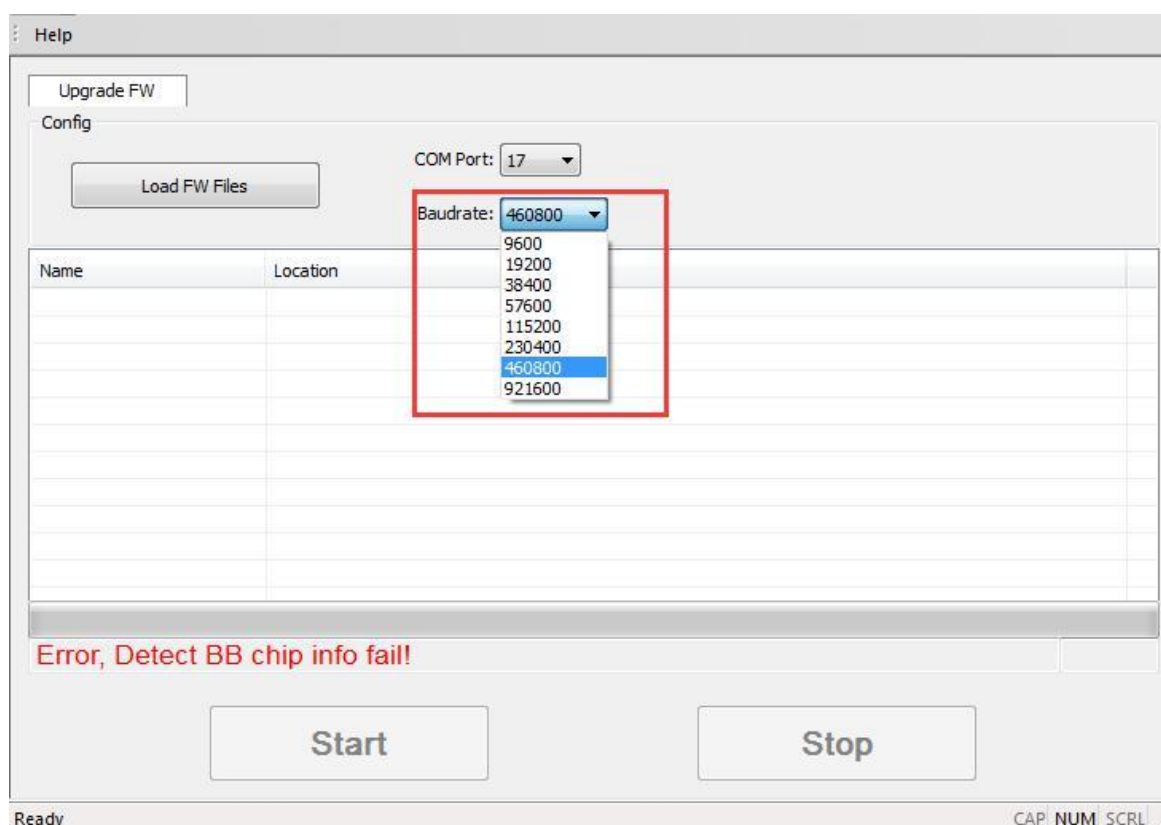
Figure 2: Select the Correct Serial Port for Mxx/GCxx/BCxx Modules

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1.1.2. Set Baud Rate

Click the “**Baud rate**” dropdown list and select an appropriate baud rate. It is recommended to select 460800 for M66 other Quested modules, as shown in the following figure.

**Figure 3: Select the Baud Rate****NOTES**

1. Baud rates have many different values, and it is the hardware environment that determines whether a specified baud rate can be supported. If not supported, an error message will be returned.

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1.2. Load Firmware Files**1.2.1. Load APP Firmware for Open CPU or Quecopen Modules**

This step is only necessary for Quectel Open CPU or Quecopen modules.

Step 1: Click the button “**Load FW Files**” and select the **.cfg** file which needs to be downloaded to the module.

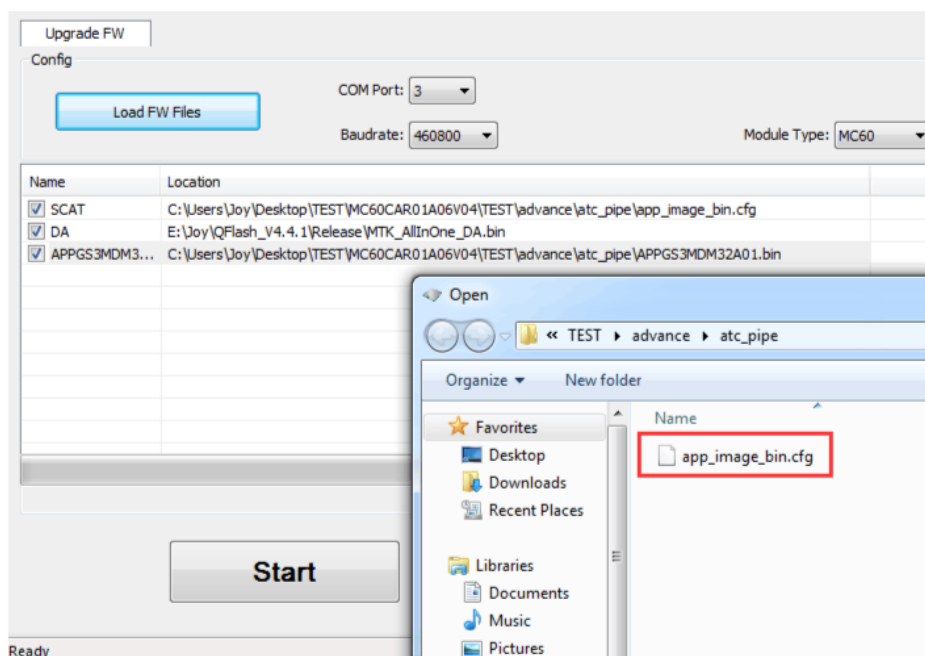


Figure 4: Select the .cfg File

NOTE

The path where the firmware is stored should NOT contain any spaces, and English characters are preferred.

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Step 2: Click the “**Module Type**” dropdown list and choose an appropriate module type. (M66)

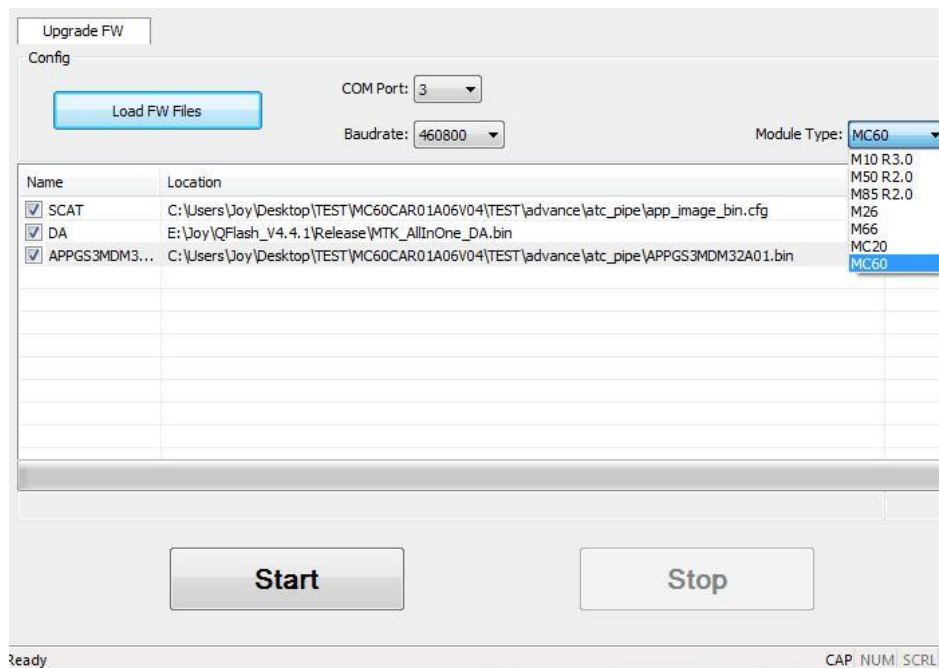


Figure 5: Select the Module Type

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1.3. Upgrade the Firmware

Step 1: before Click to “**Start**” button to upgrade the firmware disconnect module power supply. after clicking “**Start**” button then connect the module power supply. It will start to upgrade the firmware as shown in the following figure.

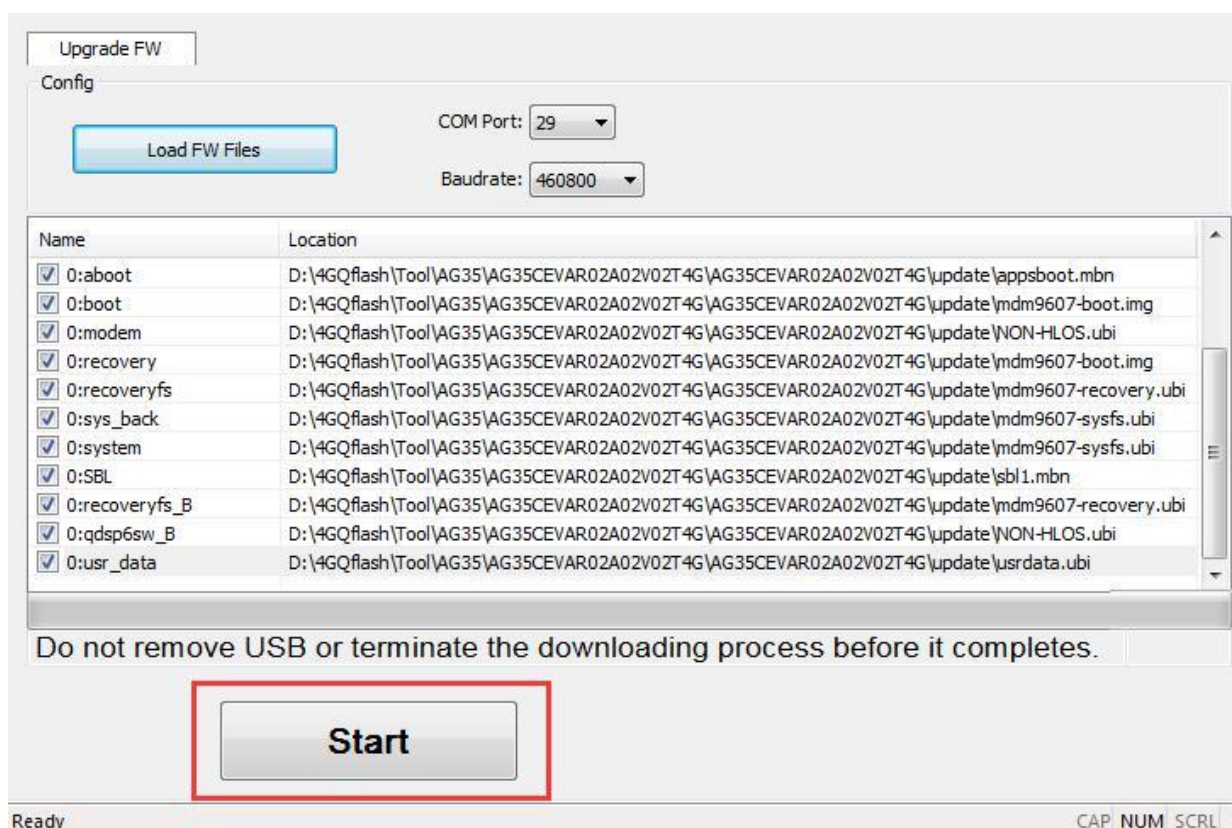


Figure 6: Click the Start Button

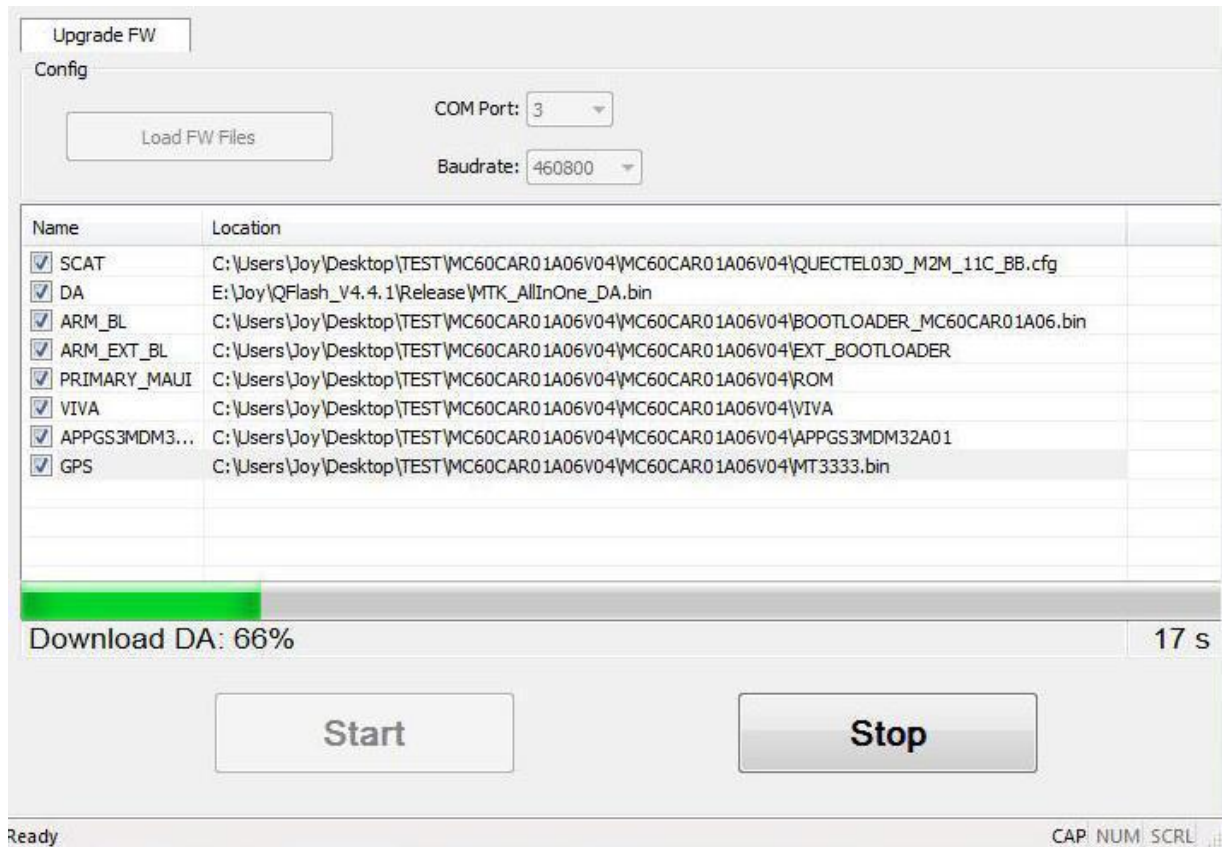
NOTES

1. Please note that it is NOT permitted to stop the upgrading process, and please do NOT remove USB or terminate the downloading process before upgrading is completed.

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Step 2: It will start to upgrade the firmware as shown in the following figure.



Upgrade FW

Config

Load FW Files

COM Port: 3

Baudrate: 460800

Name	Location
☒ SCAT	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\QUECTEL03D_M2M_11C_BB.cfg
☒ DA	E:\Joy\QFlash_V4.4.1\Release\MTK_AllInOne_DA.bin
☒ ARM_BL	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\BOOTLOADER_MC60CAR01A06.bin
☒ ARM_EXT_BL	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\EXT_BOOTLOADER
☒ PRIMARY_MAU	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\ROM
☒ VIVA	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\VIVA
☒ APPGS3MDM3...	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\APPGS3MDM32A01
☒ GPS	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\MT3333.bin

Download DA: 66% 17 s

Start Stop

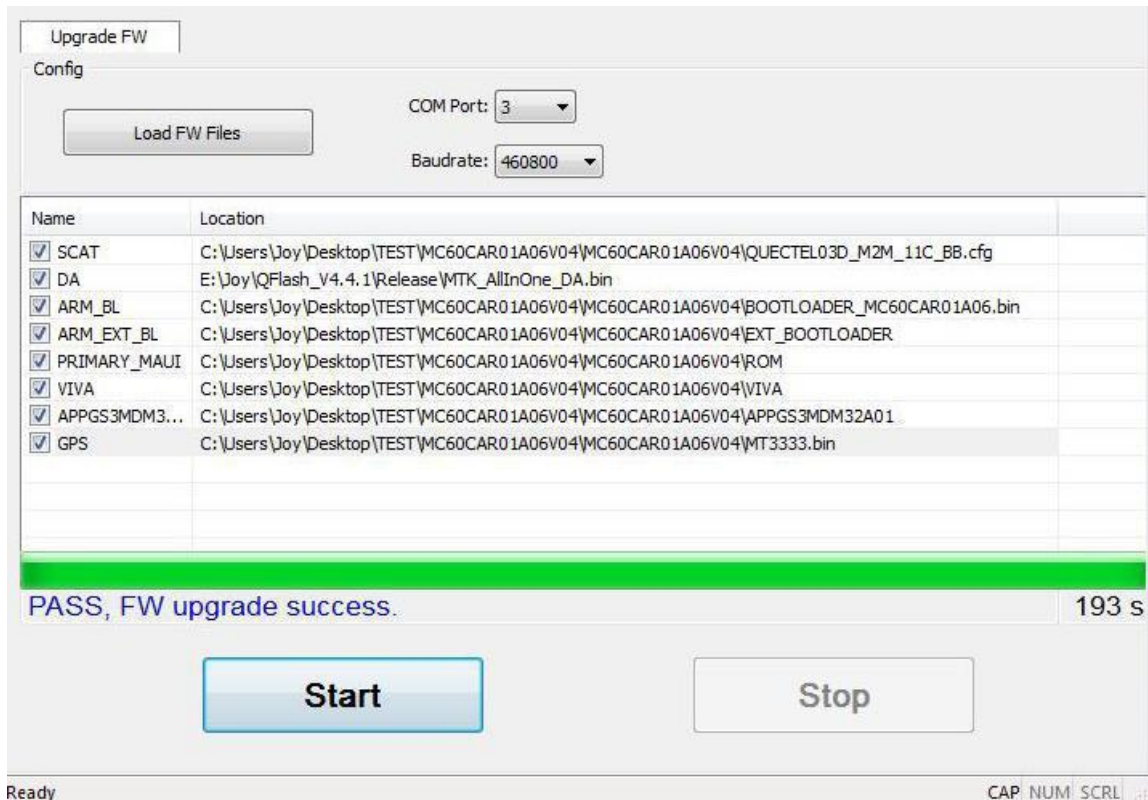
Ready CAP NUM SCRL

Figure 7: Start to Upgrade after Restarting the Mxx Modules

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Step 3: “FW upgrade success” will be shown on the interface after the firmware has been successfully upgraded, as shown in the following figure.



Upgrade FW

Config

Load FW Files

COM Port: 3

Baudrate: 460800

Name	Location
☒ SCAT	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\QUECTEL03D_M2M_11C_BB.cfg
☒ DA	E:\Joy\QFlash_V4.4.1\Release\MTK_AllInOne_DA.bin
☒ ARM_BL	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\BOOTLOADER_MC60CAR01A06.bin
☒ ARM_EXT_BL	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\EXT_BOOTLOADER
☒ PRIMARY_MAU1	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\ROM
☒ VIVA	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\VIVA
☒ APPGS3MDM32A01	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\APPGS3MDM32A01
☒ GPS	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\MT3333.bin

PASS, FW upgrade success. 193 s

Start Stop

Ready CAP NUM SCRL

Figure 8: Successful Upgrade

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1.4. Abnormalities

Abnormalities may be caused by incorrect parameter of baud rate, damaged EVB/TE-B or invalid files, etc.

The following illustrates some common abnormalities.

1.4.1. Selected a Wrong Serial Port

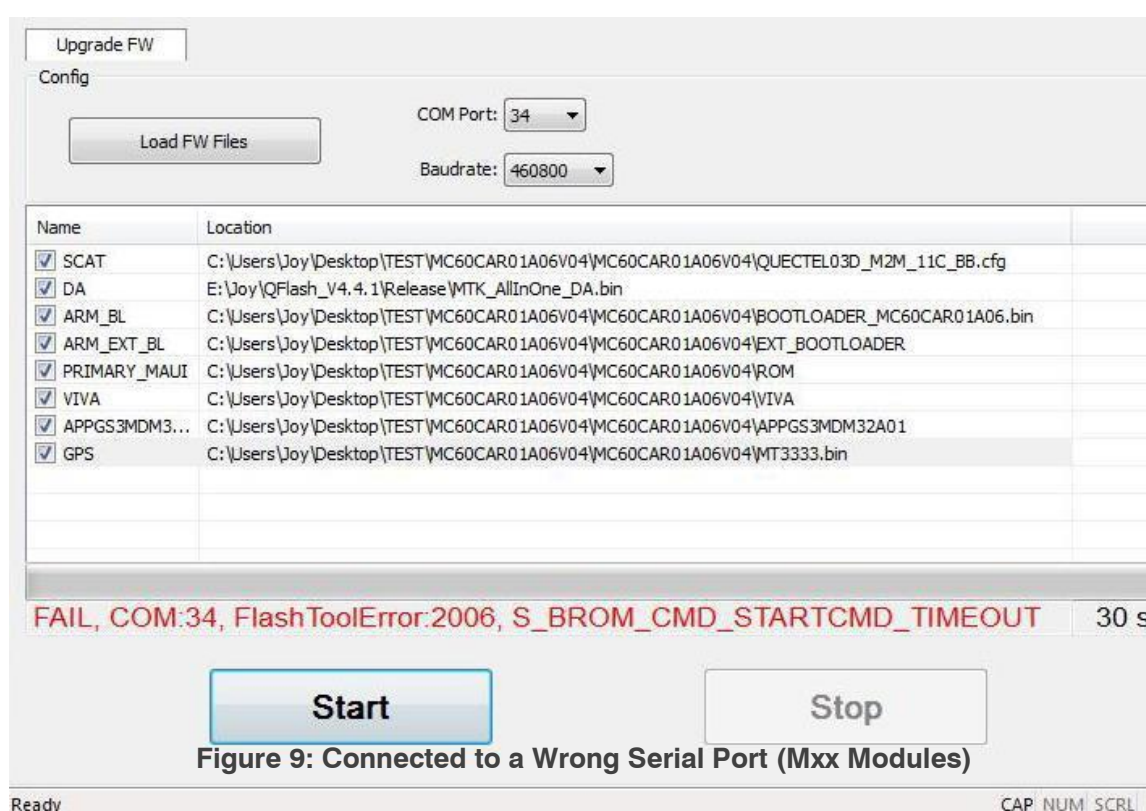


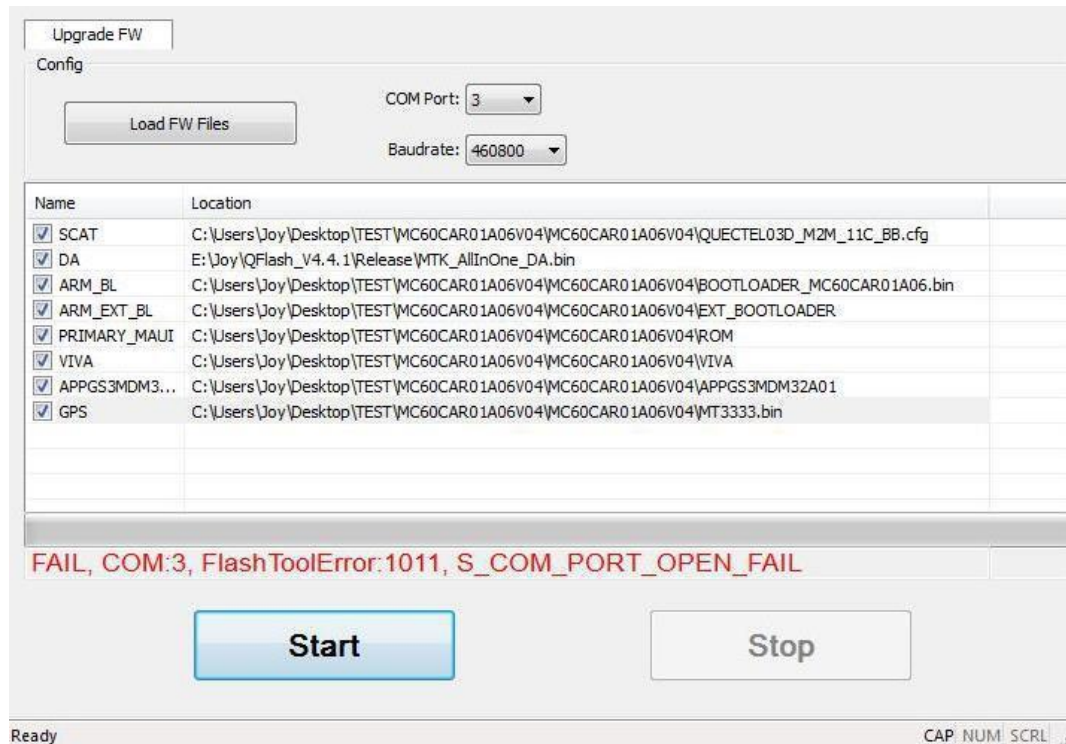
Figure 9: Connected to a Wrong Serial Port (Mxx Modules)

NOTE

After selecting a correct serial port, if the Mxx modules are not restarted, then the error message will be the same as that of selecting a wrong serial port.

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1.4.2. Connected to an Occupied Serial Port


Upgrade FW

Config

Load FW Files

COM Port: 3

Baudrate: 460800

Name	Location
☒ SCAT	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\QUECTEL03D_M2M_11C_BB.cfg
☒ DA	E:\Joy\QFlash_V4.4.1\Release\MTK_AllInOne_DA.bin
☒ ARM_BL	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\BOOTLOADER_MC60CAR01A06.bin
☒ ARM_EXT_BL	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\EXT_BOOTLOADER
☒ PRIMARY_MAUI	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\ROM
☒ VIVA	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\VIVA
☒ APPGS3MDM3...	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\APPGS3MDM32A01
☒ GPS	C:\Users\Joy\Desktop\TEST\MC60CAR01A06V04\MC60CAR01A06V04\MT3333.bin

FAIL, COM:3, FlashToolError:1011, S_COM_PORT_OPEN_FAIL

Start Stop

Ready CAP NUM SCRL

Figure 10: Connected to an Occupied Serial Port (Mxx Modules)

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1.4.3. Selected an Unsupported Baud Rate

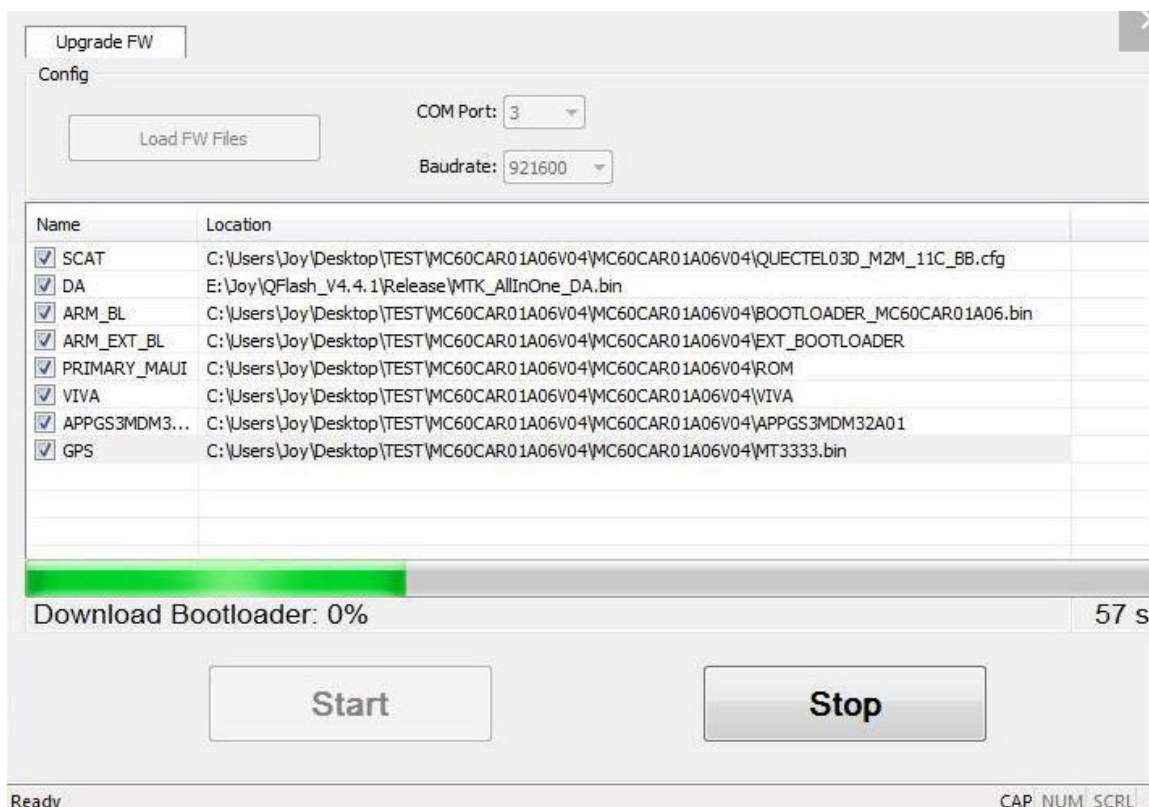


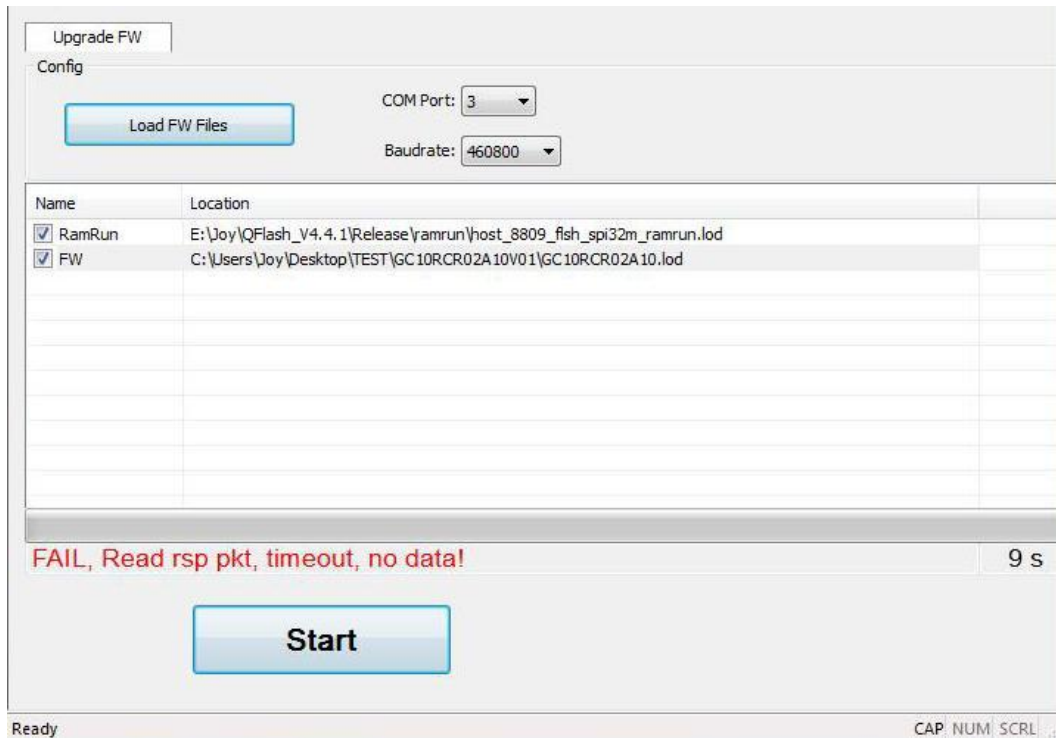
Figure 11: An Unsupported Baud Rate is Selected (Mxx Modules)

NOTE

For Mxx modules, if an unsupported baud rate is selected, the tool will stop to run and no error message will be prompted. In such case, please click the “**Stop**” button to re-select a supported baud rate to upgrade.

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1.4.4. Selected an Invalid Load File

Upgrade FW

Config

Load FW Files

COM Port: 3

Baudrate: 460800

Name	Location
☒ RamRun	E:\Joy\QFlash_V4.4.1\Release\ramrun\host_8809_fish_spi32m_ramrun.lod
☒ FW	C:\Users\Joy\Desktop\TEST\GC10RCR02A10V01\GC10RCR02A10.lod

FAIL, Read rsp pkt, timeout, no data! 9 s

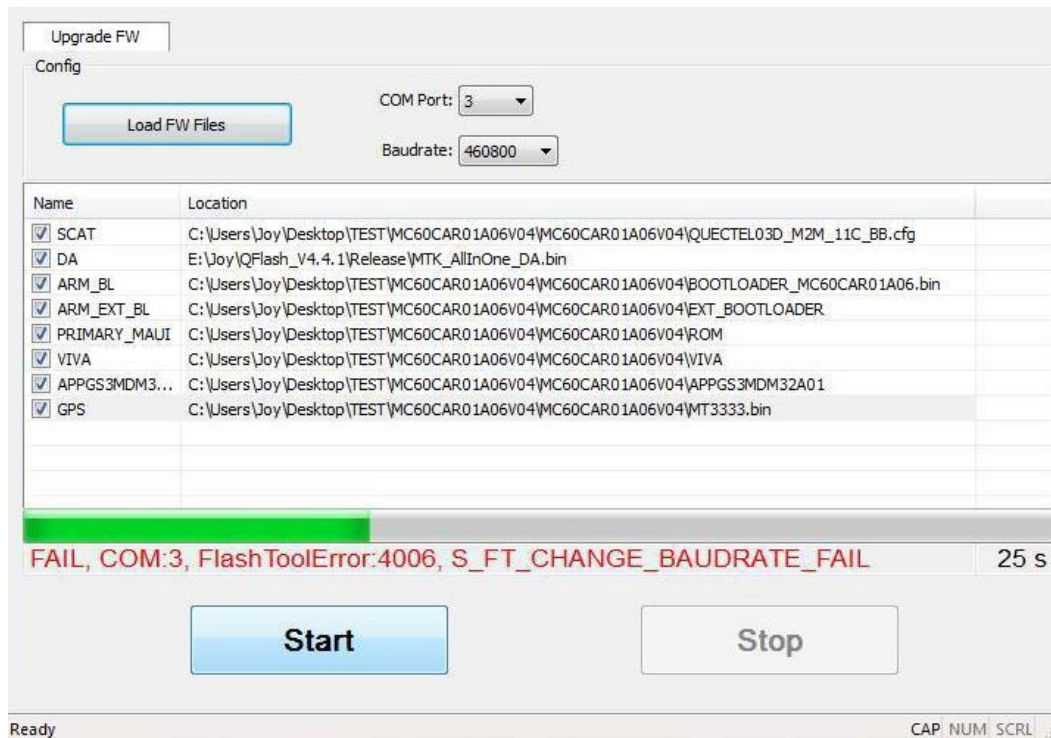
Start

Ready CAP. NUM SCRL

Figure 12: An Invalid Scatter File is Selected (Mxx Modules)

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1.4.5. Power Supply is Abnormal**Figure 13: Power Supply is Abnormal (Mxx Modules)**

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1.4.6. USB to RS-232 Converter Cable is Abnormal

Upgrade FW

Config

Load FW Files

COM Port: 29

Baudrate: 460800

Name	Location
☒ 0:aboot	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\appsboot.mbn
☒ 0:boot	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\mdm9607-boot.img
☒ 0:modem	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\NON-HLOS.ubi
☒ 0:recovery	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\mdm9607-boot.img
☒ 0:recoveryfs	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\mdm9607-recovery.ubi
☒ 0:sys_back	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\mdm9607-sysfs.ubi
☒ 0:system	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\mdm9607-sysfs.ubi
☒ 0:SBL	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\sbl1.mbn
☒ 0:recoveryfs_B	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\mdm9607-recovery.ubi
☒ 0:qdsp6sw_B	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\NON-HLOS.ubi
☒ 0:usr_data	D:\4GQflash\Tool\AG35\AG35CEVAR02A02V02T4G\AG35CEVAR02A02V02T4G\update\usrdata.ubi

FAIL, Send_Mbn_Pkt_Data_V3 timeout [Send_Mbn_Pkt_Data_V3]

37 s

Start

Ready

CAP NUM SCRL

Figure 14: USB Converter Cable is Abnormal

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Docklight User Guide

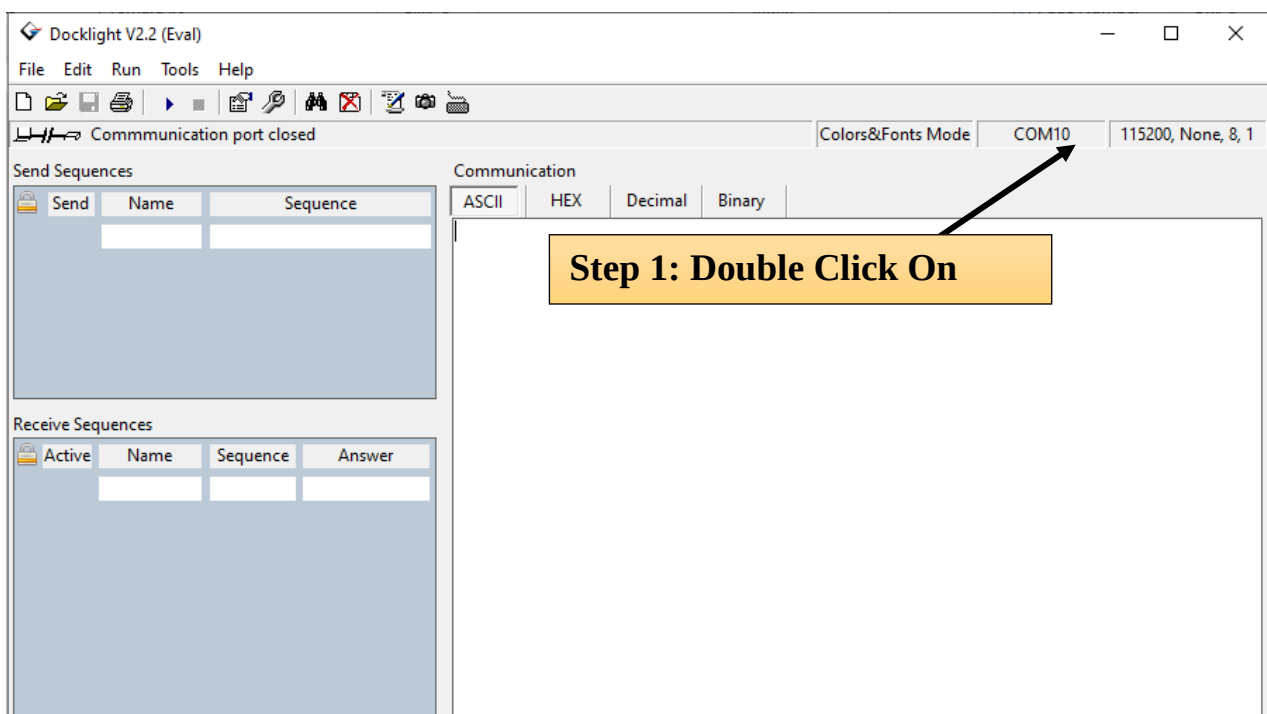
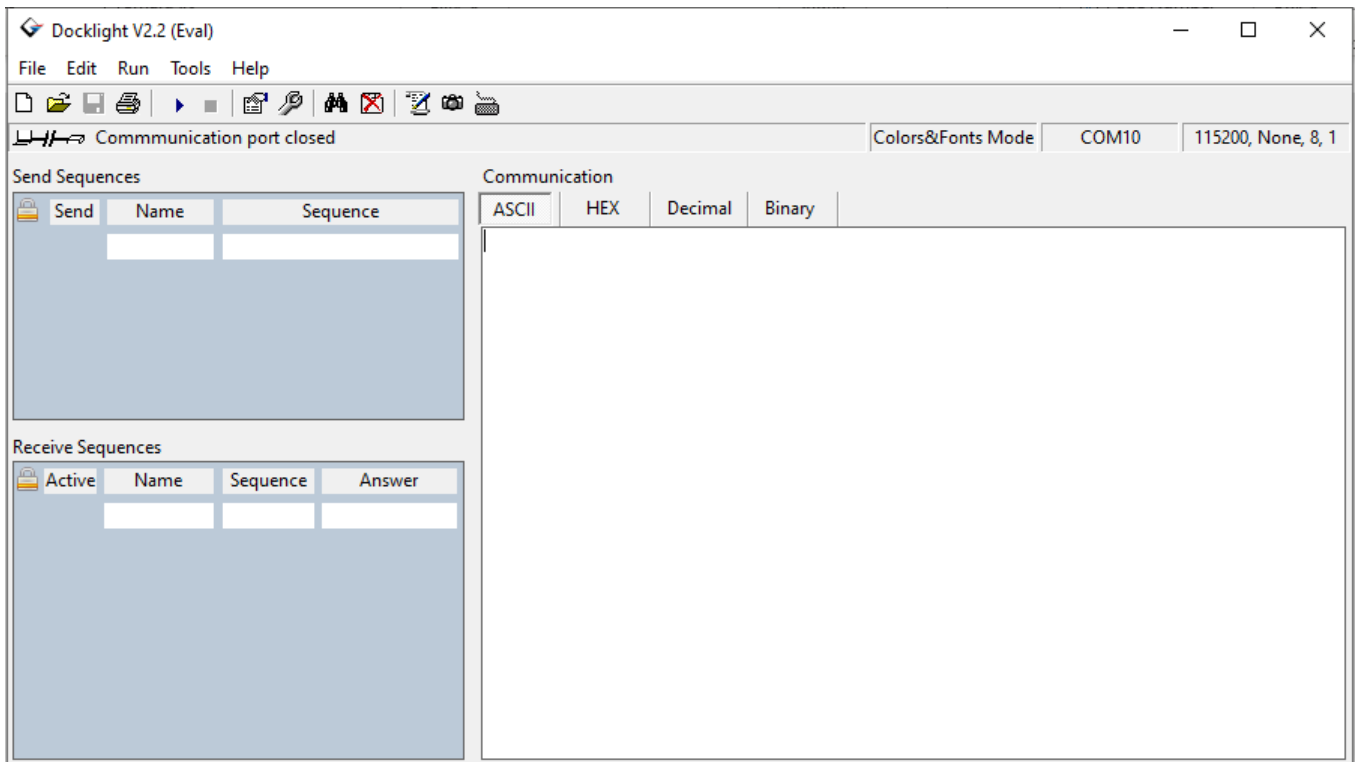
Rev. Docklight_V2.2.8

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Configure Serial Port and Baud Rate

When the docklight tool is opened, the main interface will show as below.



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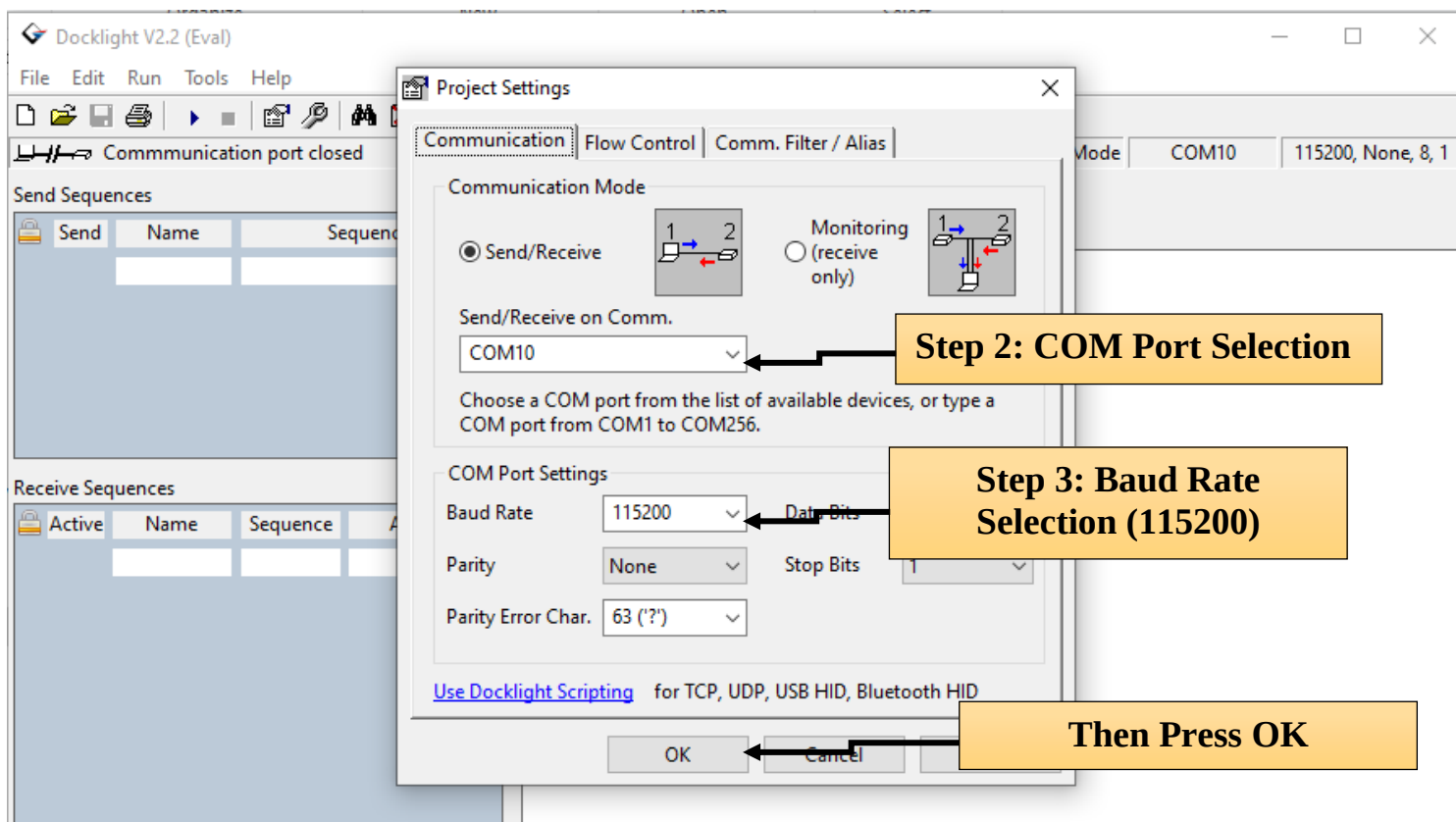
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Then,

Step 2: Select Com Port which one you are using.

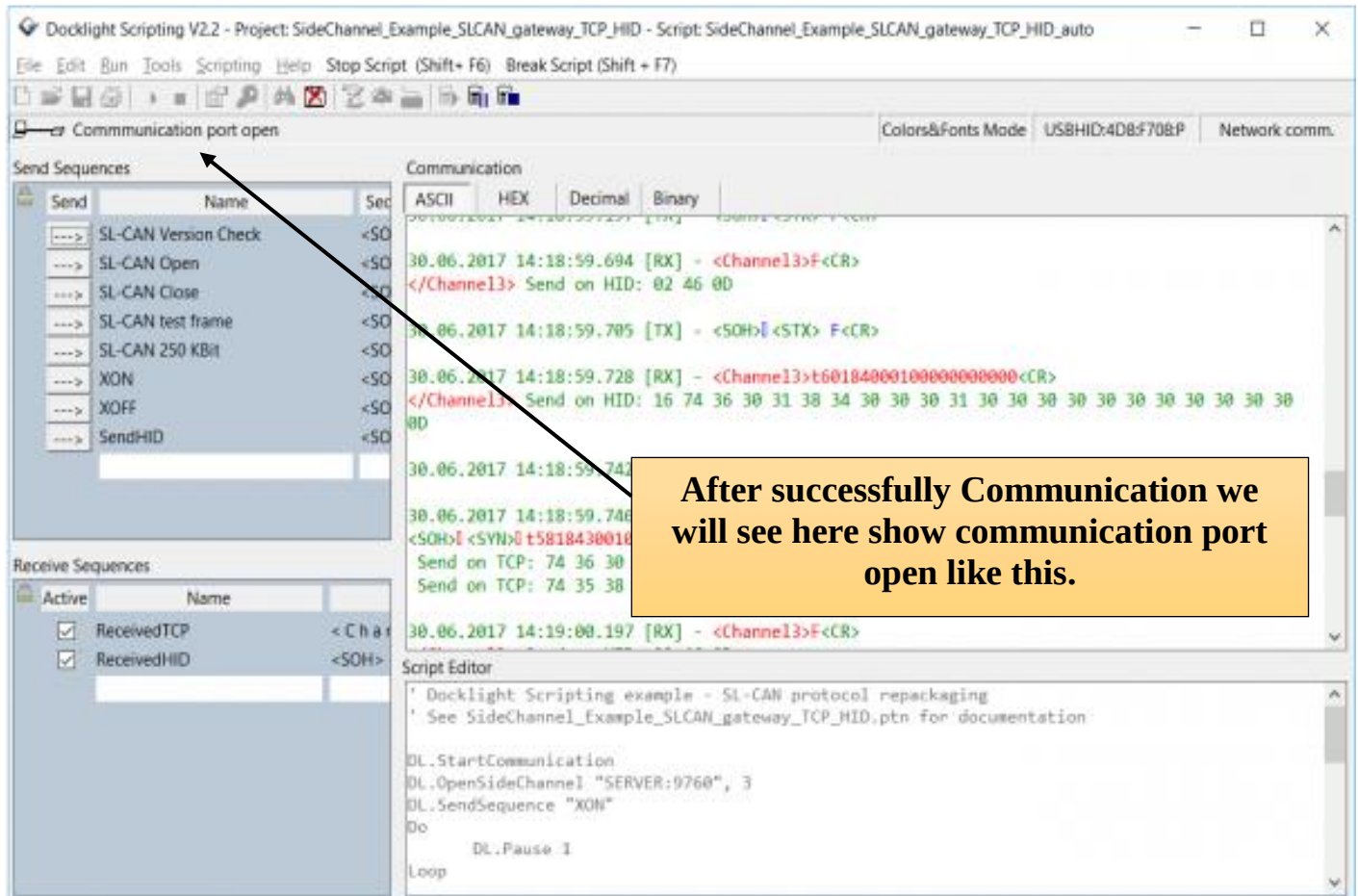
Step 3: Select Baud Rate 115200 For PC to Konnect Device.

Then press OK.



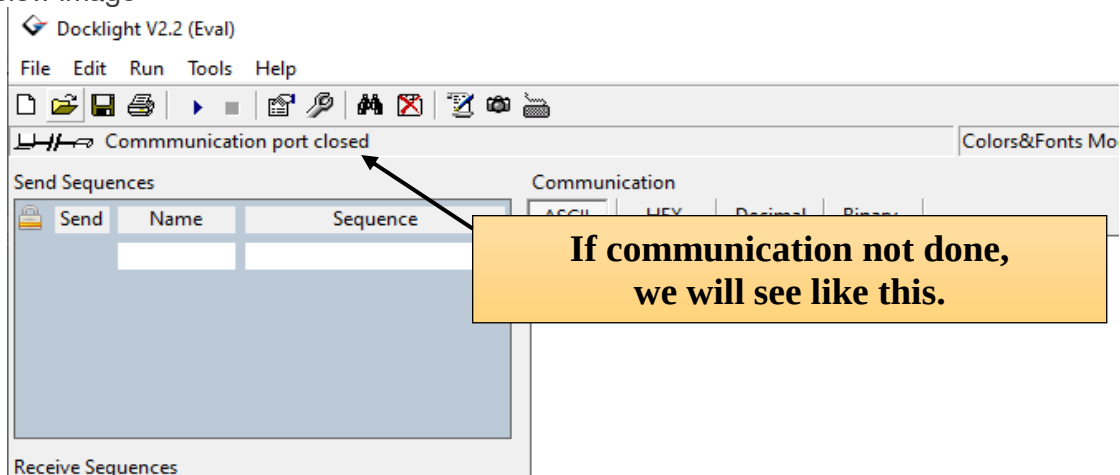
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If there is **Communication Port Closed** then communication between PC and Konnect device not done, need to check USB cables or COM port or baud rate.

Ex. See below image



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After Successfully communication we need to set device ID to Connect Device as per our Konnect identification plan.

See below Fig.

K-Connect installation plan.

Oracle\Part number	Device Identification region wise							
	Delhi	Kolkata	Gujarat	Mumbai	Pune	Bangaluru	Chennai	Hyderabad
56.137.50.411	XX - 1 - XXXXX	XX - 2 - XXXXX	XX - 3 - XXXXX	XX - 4 - XXXXX	XX - 5 - XXXXX	XX - 6 - XXXXX	XX - 7 - XXXXX	XX - 8 - XXXXX

Device Identification discription

X	X	X	X	X	X	X	X	X
Device ID Chiller Model Wise		Region Identification		Device Number settable while programing				

56.137.50.349	8	0						
56.137.50.372	8	1						
56.137.50.374	8	2						
56.137.50.375	8	3						
56.137.50.386	8	4						
56.137.50.399	8	5						

As shown Below table in Konnect protocol folders you will see below 5 folders P1,P2,P3,P4 & P5.

Folder P1	80	KWK/KWI Single Circuit -56.137.50.349
Folder P2	81	KWK /KWI Double Circuit - 56.137.50.372
Folder P3	82	KWS/KAS Single Circuit -56.137.50.374
Folder P4	83	KWS/KAS Double Circuit -
Folder P5	84	KSC Single Circuit - 56.137.50.386

For EX... If you are installing first device For KSC Single Ckt (Pune Branch).

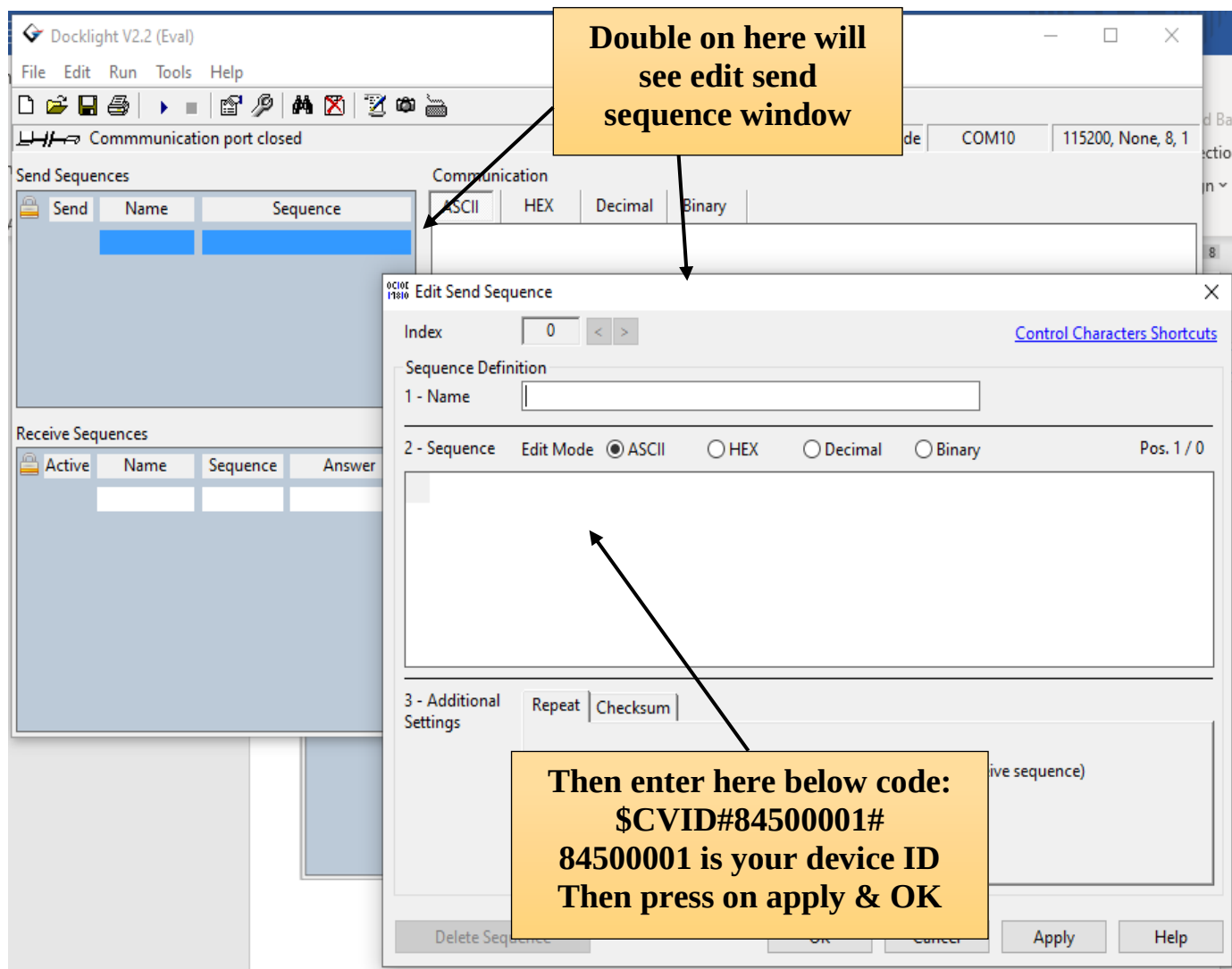
Your address will be like this.

Device ID Chiller Model Wise		Region Identification	Device Number settable while programing				
8	4	5	0	0	0	0	1

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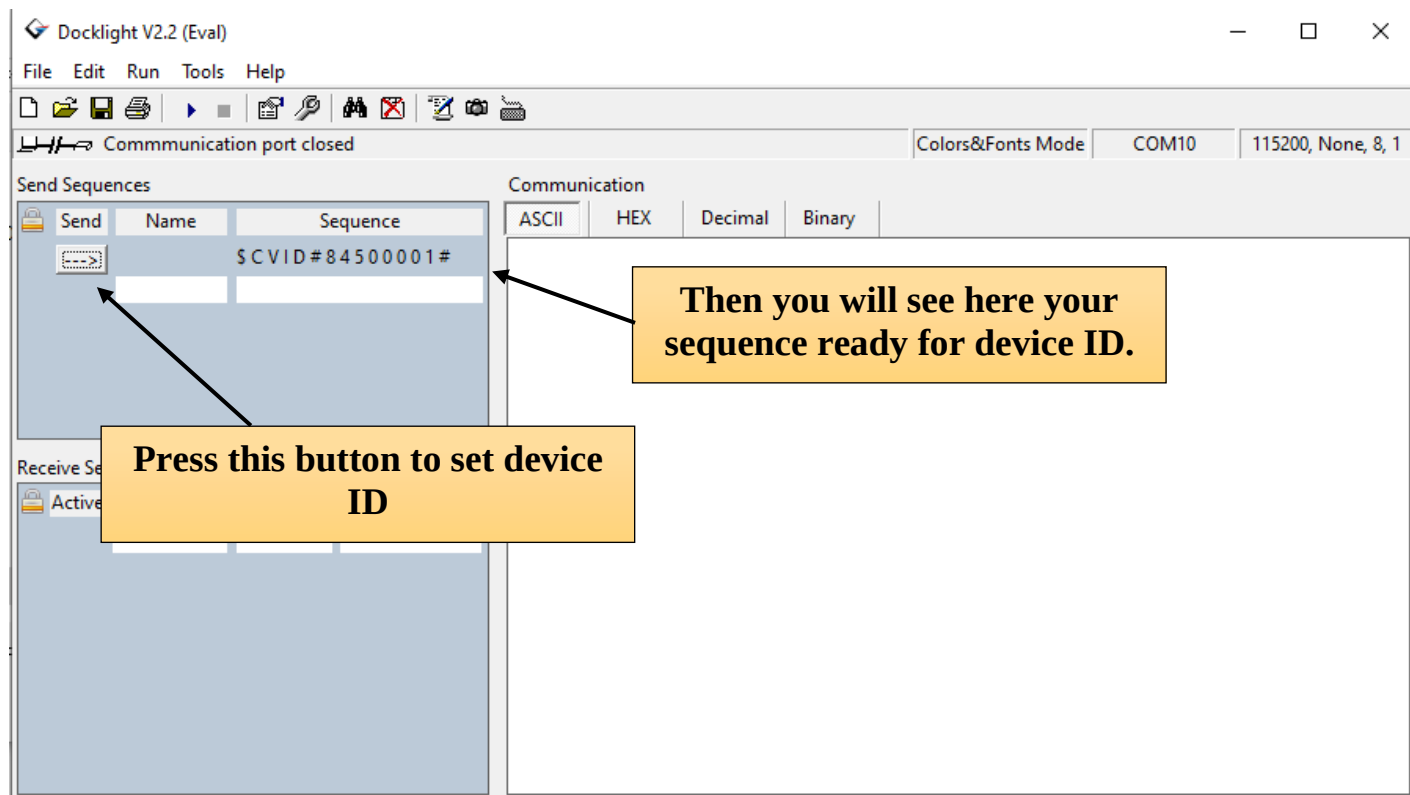
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Now after creation of device ID need to assign device id to Konnect device, as show below image,

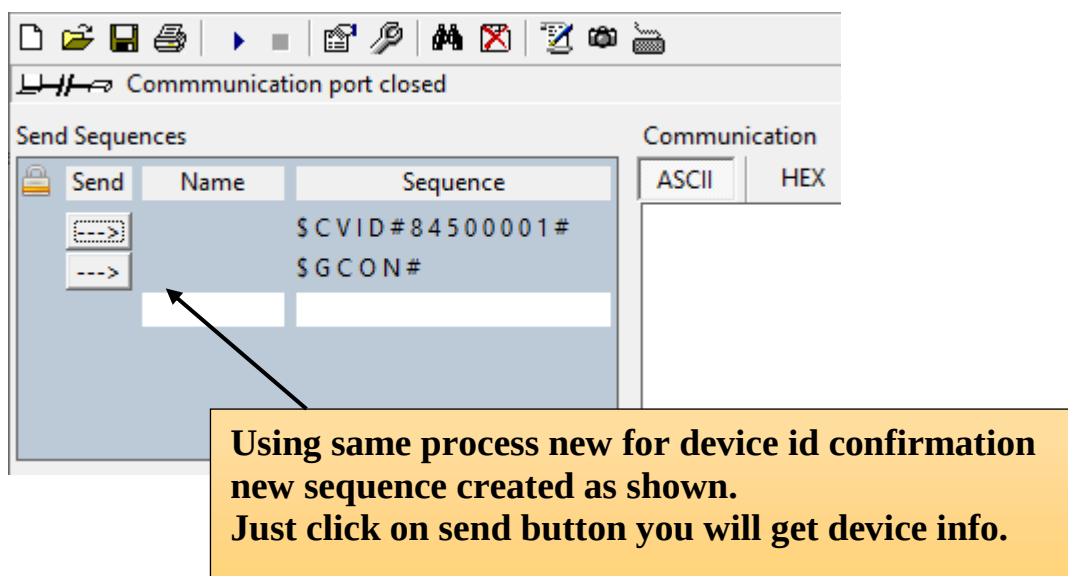


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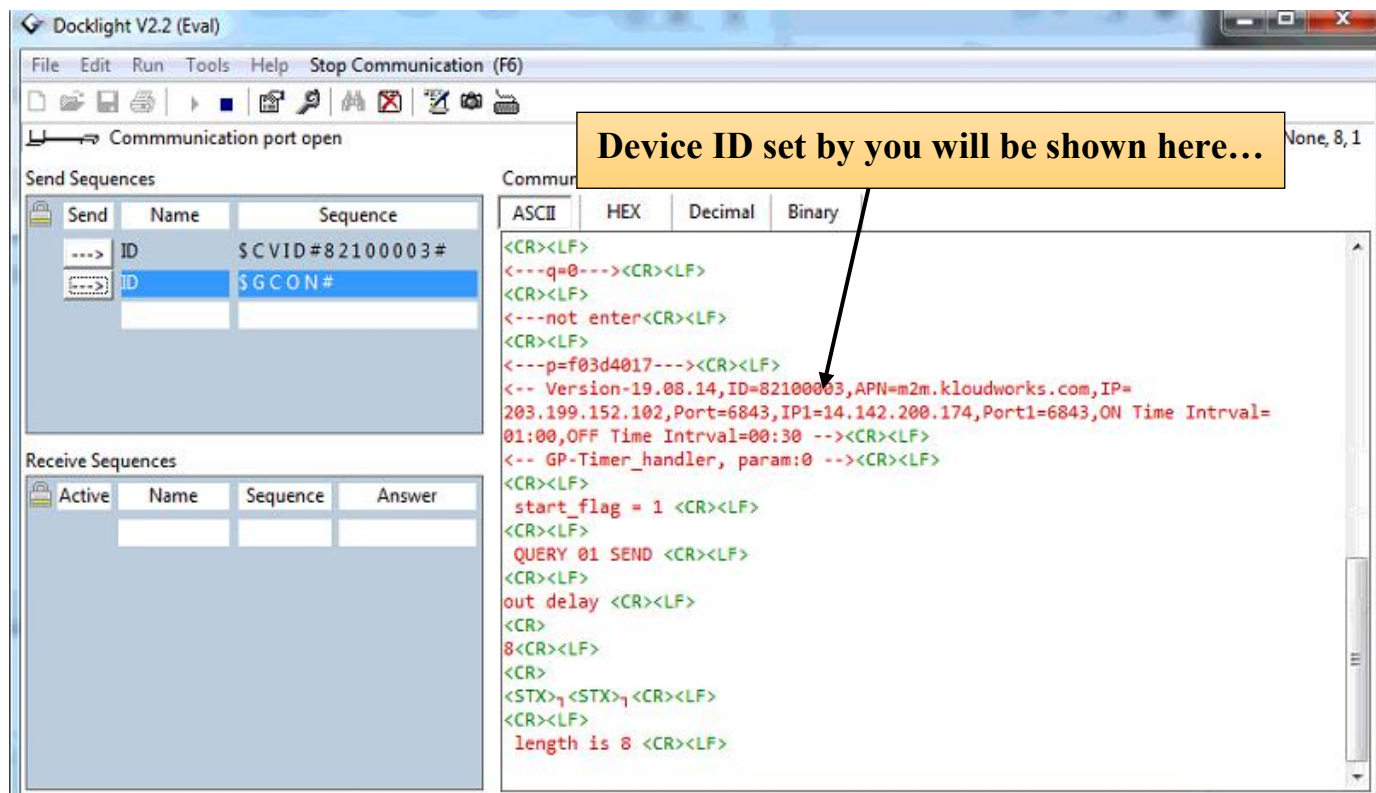
Same as per device ID create sequence for **\$GCON#** this code so you can confirm where device ID set or not.



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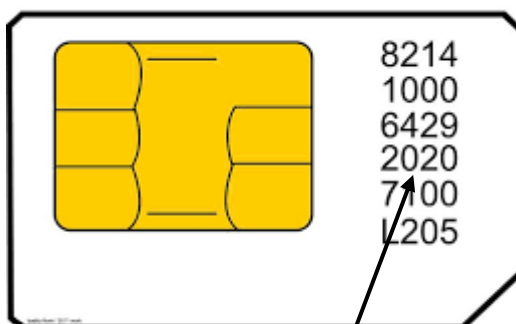
See one more example below...



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Now your device configuration completed just take sim details and site details.



Not down this number with device ID for details.

Now go for installation...